

Biomolecular Structure & Organization
Question Bank

Questions:-

1. Explain the structure of an Atom and its properties.
2. Which are the abundant elements in biomolecules?
3. What is a covalent bond?
4. What type of interactions do the atoms have with each other?
5. What are the atomic forces?
6. What are molecular orbitals?
7. How and why can atoms form chain compounds?
8. What are proteins?
9. What are amino acids?
10. What is the peptide bond?
11. What are the levels of protein structure?
12. Write a short note on Ramachandran plot.
13. What are torsional angles in proteins?
14. Differentiate between Essential and Nonessential amino acids.
15. Differentiate between Standard and Non-standard amino acids.
16. Give the properties of Alpha-helix from secondary structure.
17. Give the examples for Types of Helices.
18. Explain Beta-sheets from secondary structure.
19. Draw the structure of the Greek Key.
20. Write a short note on Tertiary Structure.
21. Give some examples of $\alpha+\beta$ topology.
22. Write a short note on Quaternary structure.
23. What is meant by Contact surface, Reentrant surface, Molecular surface and Accessible surface.
24. What is Protein Packing Density?
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